



Dynamic underwater cognition: aligned detection networks for enhanced underwater object recognition

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Abstract

Complex underwater environments, characterized by blurred objects and strong background interference, present significant challenges for existing object detection algorithms. Accurate and efficient underwater object detection is crucial for sustainable marine resource development and conservation. This study presents the Dynamic Underwater Cognition Network (DUCNet), a lightweight yet high-accuracy model specifically designed for underwater small object detection in challenging conditions. To address underwater target blurring, we propose a Dynamic Channel-Aware Convolution (DCAConv) module that enhances feature extraction by integrating channel attention mechanisms with dynamic grouping strategies. We further develop a Dynamic Lightweight Feature Pyramid Network (DyLiteFPN) that employs dynamic kernel selection to improve multi-scale feature fusion efficiency, effectively tackling the scale variation challenges inherent in underwater imagery. Additionally, an Aligned Detection Head (AlignDet) is introduced, incorporating dynamic convolutional offsets and parameter-sharing techniques to resolve classification-localization misalignment issues in underwater object detection. Comprehensive experiments conducted on the DUO and RUOD datasets demonstrate that DUCNet achieves superior detection accuracy compared to state-of-the-art methods, with particularly notable improvements in small underwater object detection. Cross-dataset evaluations further validate the model's excellent generalization capability and robustness. Extensive ablation studies confirm the effectiveness and necessity of each proposed module in enhancing overall model performance. The source code is available at <https://github.com/lzbox/DUCNet>.

Keywords Underwater object detection · Small-object detection · Dynamic feature fusion · Lightweight deep learning

1 Introduction

The oceans cover the majority of the Earth's surface, and the abundance of marine resources is gaining increasing importance in both the food and pharmaceutical industries. With the rapid advancement of underwater unmanned vehicle (UUV) technology, underwater robots have become capable of col-

lecting marine products [1]. However, reliably and efficiently recognizing marine organisms in complex underwater environments remains a substantial challenge for these robots [2]. The marine environment is extremely intricate, with organisms exhibiting significant variation in size, shape, and distribution. Such diversity creates considerable obstacles for accurate target detection and recognition [2].

Currently, underwater target identification primarily relies on sonar or optical imaging. Sonar provides long-range imaging capabilities, but its image resolution is inherently limited. By contrast, underwater optical imaging preserves richer perceptual information, such as color, shape, and texture, and offers higher resolution and improved visualization [3]. This makes it more suitable for high-precision detection tasks. Optical image-based detection methods can generally be categorized into traditional approaches and deep learning-based methods. Classical approaches rely on manually crafted features and classifiers, such as support vector machines (SVM) and Adaboost, but they perform poorly in complex back-

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grounds and occluded environments, particularly for small or camouflaged targets.

With the emergence of deep neural networks, detection techniques based on convolutional neural networks (CNNs) and Transformers have become increasingly widespread. Two-stage detectors, such as Faster R-CNN [4] and Mask R-CNN [5], achieve high accuracy and are well-suited for multi-object detection in complex scenarios, though at the cost of substantial computational overhead. In contrast, single-stage detectors, such as YOLO [6, 7] and SSD [8], are more appropriate for real-time applications, including underwater robotics and aquaculture monitoring. Recently, Transformer-based architectures, such as DETR [9] and RT-DETR [10], have been introduced into underwater detection tasks. These models enhance background modeling through global attention mechanisms, yet their high computational demands hinder practical deployment.

Although these algorithms perform effectively in terrestrial environments, their application underwater faces persistent challenges, including motion blur, low contrast, background interference, target scale variation, and particularly poor recognition of small objects. Consequently, developing strategies for small-target detection has emerged as a critical direction in underwater vision research.

Motivated by the above challenges and technical constraints, this study introduces DUCNet, a lightweight yet high-accuracy detection model designed for small-object detection in complex underwater environments. DUCNet addresses key issues such as underwater blurring, multi-scale feature fusion, and classification-localization conflicts. During the experimental phase, images are preprocessed through cropping, rotation, and stitching. Ablation experiments confirm the effectiveness of each component, and comparative evaluations against existing networks highlight the superior performance of DUCNet. The main contributions of this study can be summarized as follows:

- **Dynamic Underwater Cognition Network (DUCNet):** Proposed a lightweight and high-accuracy underwater small-object identification system that includes DCA-Conv, DyLiteFPN, and AlignDet. DUCNet effectively addresses underwater-specific challenges such as motion blur, scale variation, and background complexity, achieving superior performance and generalization across diverse underwater datasets.
- **Dynamic Channel-Aware Convolution(DCAConv):** Proposed a dynamic channel-aware convolution module integrating channel attention, dynamic grouping, and residual connections with SiLU activation to boost sensitivity to textures and edges in blurred underwater photos.
- **Dynamic Lightweight Feature Pyramid Network(DyLite FPN):** Designed a lightweight feature pyramid network with multi-branch fusion, dynamic kernel selection,

and efficient up-sampling to improve multi-scale feature fusion and computational efficiency for small-object recognition.

- **Aligned Detection Head (AlignDet):** Introduced an aligned detection head with parameter sharing, dynamic convolutional offsets, and interactive feature modeling to optimize classification and localization tasks, enhancing accuracy in difficult underwater environments.

The remainder of this paper is structured as follows: Chapter 2, “Related Work,” systematically reviews the latest advances in underwater small-target detection. Chapter 3, “Methodology,” elaborates on the design principles and implementation details of the DUCNet model. Chapter 4, “Experiments,” comprehensively analyzes the model’s performance. Chapter 5, “Conclusion,” summarizes the research findings and discusses directions for future work.

2 Related work

2.1 Underwater object detection

Traditional methods for underwater object detection primarily rely on handcrafted features and image preprocessing techniques. These approaches generally extract attributes such as color, texture, and shape from underwater images, followed by classification using conventional machine learning classifiers [11]. For example, Chuang et al. employed phase Fourier transform to identify salient textures in fish images [11], while other studies applied saliency detection and color correction to enhance low-contrast underwater photographs. Although straightforward, such methods are highly sensitive to background clutter, occlusion, and camouflage, and they frequently fail to detect small or distant objects due to the limited representational capacity and poor generalization of handcrafted features.

In contrast, deep learning-based algorithms-particularly the YOLO (You Only Look Once) series-have become mainstream owing to their superior accuracy and real-time performance. Recent YOLO variants have been extensively adapted for underwater object detection through architectural innovations and attention mechanisms. For instance, Chen et al. [12] proposed Dynamic YOLO, which integrates deformable convolutions with unified attention fusion to enhance small-object detection; Feng et al. [13] introduced CEH-YOLO, incorporating high-order attention mechanisms; Yang et al. [14] developed UGC-YOLO, which leverages global context modeling and multi-scale fusion; and Zhang et al. [15] presented a lightweight YOLOv4-MobileNet v2 model, striking a balance between accuracy and inference speed.

Beyond CNN-based architectures, Transformer-based methods have also been explored for underwater perception.

For example, EAPT enhances semantic representation in complex underwater scenarios by integrating deformable attention with global encoder-decoder communication [16]. These approaches offer promising alternatives for underwater detection; however, challenges such as image degradation, class imbalance, and environmental noise remain significant obstacles to robust and reliable deployment.

2.2 Challenges for underwater small-object detection

Despite the remarkable progress achieved by deep learning in underwater object detection, numerous obstacles persist in practical applications, particularly in the recognition of small objects [17]. Underwater images often suffer from severe quality degradation due to light absorption and scattering, which results in low contrast, blurred details, and color distortion. Such degradation reduces the distinction between targets and background, thereby substantially increasing detection difficulty [18]. To mitigate these effects, feature enhancement techniques—particularly those incorporating spatial attention—have proven effective in emphasizing haze-relevant structures and suppressing noise, as demonstrated by FSAD-Net [19]. Furthermore, small objects frequently occupy only a few pixels, causing their features to be lost during downsampling and leading to low recall and poor detection performance.

Another critical limitation lies in the scarcity and limited generalization of underwater datasets. Deep learning algorithms depend heavily on large-scale annotated data, yet acquiring high-quality, diverse underwater imagery is both expensive and time-consuming [20]. Moreover, environmental heterogeneity across marine regions induces distribution shifts, thereby reducing model adaptability to unseen conditions. To address these challenges, some studies have explored data augmentation and generative adversarial networks (GANs) to expand training samples and improve model robustness [21]. Beyond conventional augmentation, recent cross-modal approaches such as UnderwaterImage2IR have proposed generating acoustic representations directly from visual inputs using conditional generative networks. Such cross-modal synthesis enriches dataset diversity and environmental context, offering a promising pathway for training under limited underwater data conditions [22].

Finally, the underwater environment introduces dynamic and unpredictable interferences, including suspended particles, bubbles, and light flares. These disturbances can cause partial occlusion or false positives, thereby reducing detection accuracy. Although strategies such as attention mechanisms and refined candidate filtering have been proposed to enhance resilience [24], achieving reliable detection in highly complex underwater conditions remains a fundamental and ongoing research challenge.

2.3 Relation to prior work

Existing studies have explored dynamic convolution, feature pyramid optimization, and task-aligned detection heads to address challenges in complex visual environments [25]. While these approaches provide valuable foundations, our work advances them with novel designs specifically tailored to the unique difficulties of underwater small-object detection.

Self-Calibrated Convolution (SCConv) enhances receptive fields through self-calibration, and Conditionally Parameterized Convolution (CondConv) adapts kernel parameters based on input conditions. However, these methods either rely on static grouping strategies or introduce significant computational overhead [26]. To overcome these limitations, we propose the Dynamic Channel-Aware Convolution (DCAConv) module, which integrates dynamic grouping with channel attention. This design enables the model to adaptively emphasize texture and edge information while maintaining lightweight efficiency. As a result, DCAConv strengthens feature extraction for blurred underwater targets and consistently improves detection accuracy.

Feature pyramid networks have also been extensively studied for multi-scale integration. For example, the Bi-directional Feature Pyramid Network (BiFPN) achieves effective cross-scale fusion by assigning adaptive weights [27, 28], while the neck structure in YOLOv8-n emphasizes efficiency through simplified fusion layers. Building upon these concepts, we introduce the Dynamic Lightweight Feature Pyramid Network (DyLiteFPN), which incorporates heterogeneous kernel selection and a CSP-MSCB block to achieve flexible multi-scale fusion. This design results in a compact architecture with fewer parameters than BiFPN, while maintaining or even improving accuracy in underwater detection tasks.

In parallel, advances in detection heads have demonstrated the value of spatial adaptability and task alignment. Deformable convolutional heads enhance localization by adapting receptive fields to irregular object shapes [29], while task-aligned heads alleviate classification-localization conflicts by refining branch interactions. Extending these concepts, our Aligned Detection Head (AlignDet) combines parameter sharing with dynamically generated offsets to achieve efficient task alignment. This design enhances recall in challenging scenarios characterized by complex lighting and scale variation, while preserving real-time inference performance.

Rather than introducing isolated modifications, this study emphasizes the complementary integration of three components: DCAConv for dynamic feature enhancement, DyLiteFPN for efficient multi-scale fusion, and AlignDet for robust task alignment. Experimental analyses and ablation studies confirm that the synergy of these modules consistently yields

substantial performance gains, particularly in small-object detection, while retaining the lightweight and fast characteristics required for underwater robotic applications.

3 Method

This section presents the target detection methodology developed in this study for complex underwater environments (as illustrated in Figure 1). At a high level, the proposed detection pipeline can be summarized as follows: given an input underwater image, the backbone first extracts multi-scale features; these features are then enhanced by the Dynamic Channel-Aware Convolution (DCAConv) module, which emphasizes discriminative channels while suppressing noise. The refined features are subsequently passed into the Dynamic Lightweight Feature Pyramid Network (DyLiteFPN), where multi-scale information is efficiently fused through lightweight and adaptive strategies. Finally, the fused features are processed by the Aligned Detection Head (AlignDet), which jointly optimizes classification and localization to produce robust detection results across varying scales. This sequential process progressively strengthens feature representation, improves cross-scale integration, and achieves a balance between detection sensitivity and accuracy.

The core architecture is inspired by the YOLO11 optimization concept and consists of three primary components: DCAConv, DyLiteFPN, and AlignDet. These modules represent the central innovations of the proposed underwater target detection framework. Specifically, the model leverages a multi-scale feature extraction backbone to capture diverse representations, enhances feature distinctiveness through DCAConv, and improves integration efficiency using a lightweight fusion mechanism. The DCAConv module significantly enhances the model's ability to detect hazy underwater targets and multi-scale features by incorporating a dynamic channel attention mechanism and an adaptive grouping strategy. The DyLiteFPN module optimizes multi-scale feature integration via a global heterogeneous kernel selection mechanism combined with weighted feature fusion, while the AlignDet module improves synergy between classification and localization through task-interactive features and dynamic convolution mechanisms. Collectively, these components enable the model to deliver multi-scale detection results that strike an effective balance between sensitivity to small objects and accurate localization of larger targets. A complete layer-by-layer specification of DUCNet is provided in Appendix A. Network architecture to facilitate reproducibility and implementation.

The remainder of this chapter is organized as follows: Section 3.1 provides a detailed discussion of the dynamic channel-aware and adaptive grouping design of the DCA-

Conv module, emphasizing its advantages in underwater object detection. Section 3.2 examines the global heterogeneous kernel selection strategy and multi-scale fusion techniques incorporated into the DyLiteFPN module. Section 3.3 explores the synergistic optimization of dynamic convolution and feature interaction in AlignDet, focusing on its role in task alignment.

3.1 DCAConv

DCAConv is a dynamic channel-aware convolution module developed from the classic DualConv module, intended to address basic challenges in underwater environments, such as target feature blurring, limited multi-scale perception, and background noise interference (as demonstrated in Figure 2). The original DualConv collects multi-scale features by parallel group convolution (Group Convolution) and pointwise convolution (Pointwise Convolution). However, its static grouping technique is difficult to adjust to the feature interaction requirements of varied channel granularities and lacks the capacity to calibrate important channels (e.g., target edges, textures), which leads target features to be overrun by background noise. Additionally, the absence of normalization and activation components limits the model's adaptability to complicated deterioration. To address these issues, DCAConv introduces three core improvements: First, it embeds a channel attention mechanism, dynamically weighting key channels through global average pooling and fully connected layers to enhance the response strength of discriminative features such as textures and edges; second, it adopts a dynamic grouping strategy, adaptively adjusting the grouping granularity based on the number of input channels, optimizing the fusion efficiency of local and global features for targets of different scales; and third, it introduces residual connections and the SiLU activation function to mitigate the gradient vanishing problem in deep networks and improve the model's ability to express low-contrast features. This module provides a more effective feature extraction approach for underwater target identification through dynamic perception and adaptive fusion processes.

The original DualConv merges features using simple convolution processes. The enhanced version incorporates batch normalization (BatchNorm) and the SiLU activation function to boost nonlinear expressiveness and relieve the gradient vanishing problem. To ensure compatibility between branches, both the grouped convolution and the pointwise convolution are designed to output the same channel dimension C_{out} , so that their results can be added directly. The mathematical expression of the enhanced DualConv is as

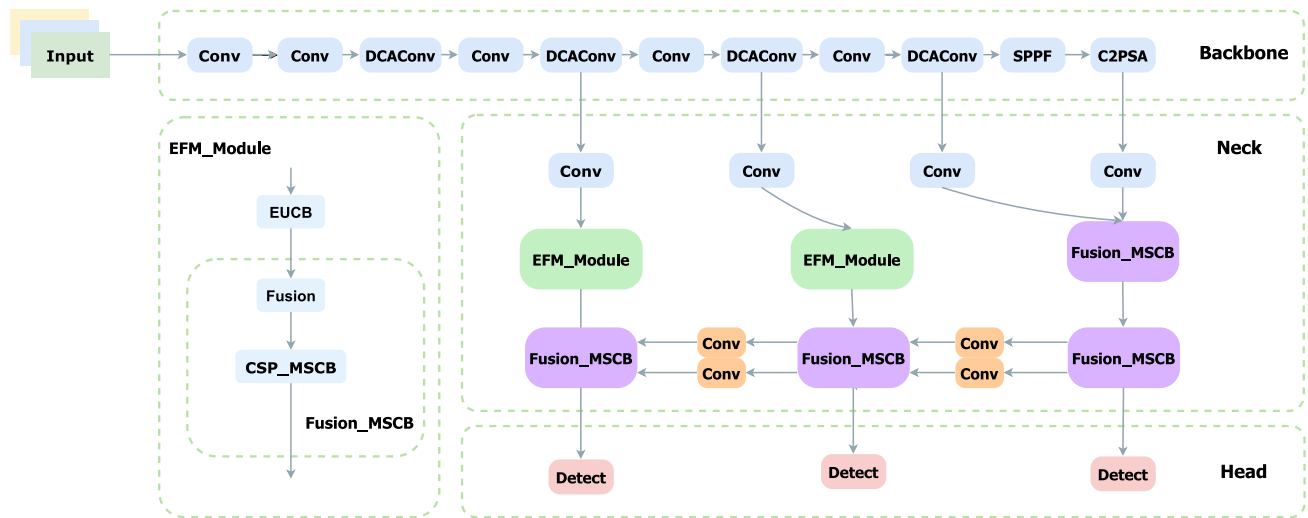


Fig. 1 Structure of DUCNet

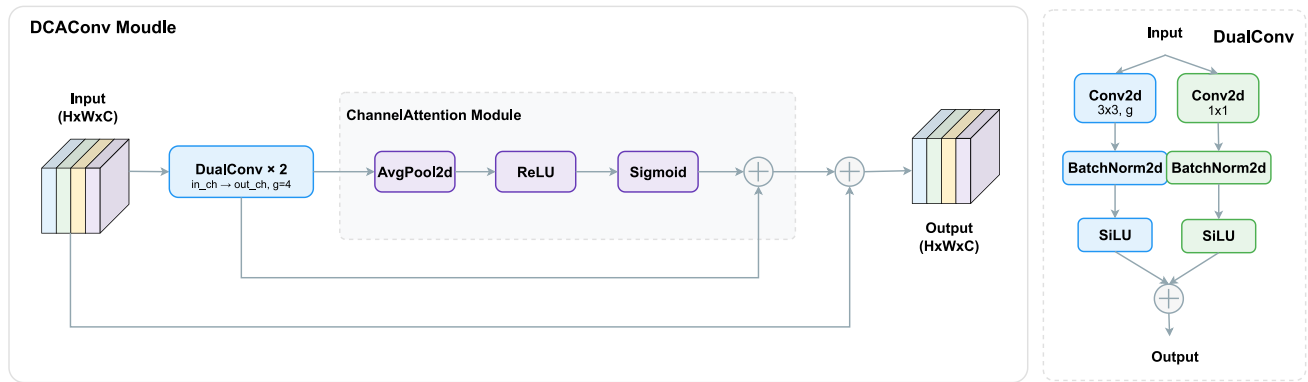


Fig. 2 Structure of DCACConv module

follows:

$$\begin{cases} F_{gc} = \text{SiLU}(\text{BN}(\text{Conv}_{3 \times 3}(F_{in}))) \\ F_{pwc} = \text{SiLU}(\text{BN}(\text{Conv}_{1 \times 1}(F_{in}))) \\ F_{out} = F_{gc} + F_{pwc} \end{cases} \quad (1)$$

where F_{in} denotes the input feature, $\text{Conv}k \times k$ represents the convolution operation with a kernel size of $k \times k$, BN indicates batch normalization, SiLU denotes the Sigmoid Linear Unit activation function, F_{gc} refers to the output feature of group convolution, F_{pwc} denotes the output feature of pointwise convolution, and F_{out} represents the final output feature.

The proposed DCACConv module expands the normal DualConv block by including consecutive dual convolution operations, a channel attention mechanism, and a residual connection. This design increases feature expression and simplifies gradient propagation, which is particularly helpful for deep designs in complicated underwater settings. Given the input feature F_{in} , the computing method within the DCA-

Conv module is formulated as:

$$\begin{cases} F_{dual1} = \text{DualConv}(F_{in}) \\ F_{dual2} = \text{DualConv}(F_{dual1}) \\ F_{out} = F_{in} + F_{dual2} \odot CA(F_{dual2}) \end{cases} \quad (2)$$

where $\odot$ denotes element-wise (Hadamard) multiplication across channels and spatial locations. Two layered DualConv procedures progressively enhance the input characteristics. The output of the second DualConv, F_{dual2} , is recalibrated using a channel attention (CA) process that gives adaptive weights to each channel. The reweighted features are then merged with the original input through a residual connection. This topology enables the network to retain low-level spatial information from F_{in} while boosting the semantic discriminability of deeper features.

The channel attention mechanism dynamically calibrates the relevance weights of channels using global average pooling and fully linked layers. Given the input feature $F \in$

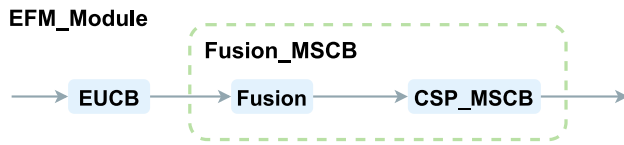


Fig. 3 Structure of EFM_Module

$\mathbb{R}^{B \times C \times H \times W}$, the attention weights are computed as follows:

$$w_c = \sigma(\mathbf{W}_2 \cdot \text{ReLU}(\mathbf{W}_1 \cdot \text{GAP}(F))) \quad (3)$$

where GAP denotes global average pooling, $\mathbf{W}_1 \in \mathbb{R}^{C/r \times C}$, $\mathbf{W}_2 \in \mathbb{R}^{C \times C/r}$ are the parameters of the fully connected layers, and $r = 16$ is the compression ratio. The final output feature is $F_{\text{att}} = F \otimes w_c$, where $\otimes$ represents channel-wise multiplication. This technique assigns higher weights to discriminative channels (such as textures and edges) for underwater target recognition.

To ease the degradation problem in deep networks, DCA-Conv integrates a residual connection technique, which permits more effective gradient backpropagation. The number of groups g in DCAConv is dynamically determined dependent on the number of input channels:

$$g = \max\left(4, \left\lfloor \frac{C_{\text{in}}}{16} \right\rfloor\right) \quad (4)$$

where C_{in} denotes the number of input channels and $\lfloor \cdot \rfloor$ represents the floor operation. In our setting, $C_{\text{in}} = 128$, which yields $g = \lfloor 128/16 \rfloor = 8$. This dynamic grouping rule ensures that layers with a larger number of channels are assigned a finer-grained partition, thereby enhancing local feature interactions, while layers with fewer channels are prevented from being excessively fragmented by enforcing a lower bound of $g = 4$. The detailed influence of different group numbers g on detection performance is further analyzed in the ablation study section, where we demonstrate that this dynamic rule provides a balanced trade-off between precision and recall.

Through this dynamic grouping mechanism, the DCA-Conv module not only maintains robustness across different channel dimensions but also adaptively exploits the optimal granularity of feature interactions, which is crucial for extracting discriminative underwater features under conditions of blur, noise, and scale variation.

3.2 DyLiteFPN

The detection of small underwater targets faces various problems, including low contrast, picture blur, complicated backgrounds, and target scale change. Traditional feature pyramid networks (FPNs) boost detection performance by fusing multi-scale features; however, their fixed receptive

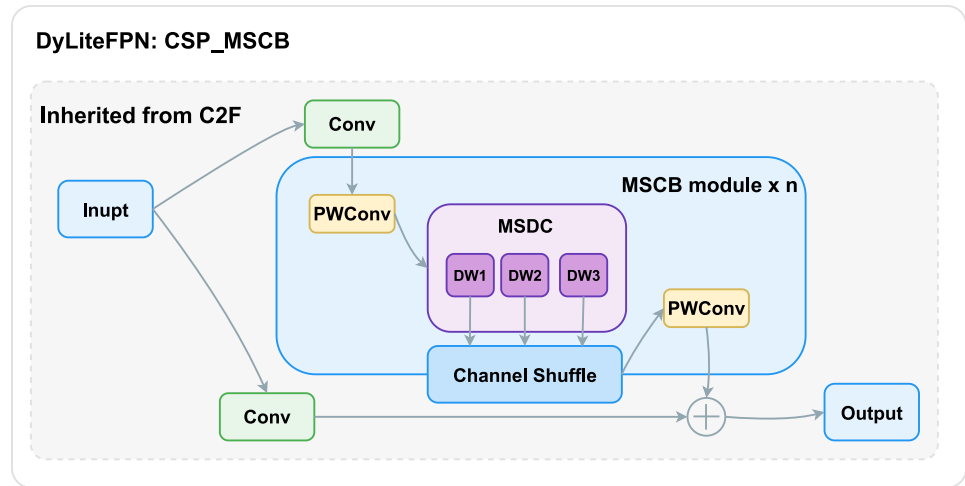
fields and simple fusion algorithms generally fall short in underwater situations. Specifically, conventional FPNs cannot dynamically adjust to different target scales, which results to the loss of important characteristics for small targets or spatial compression for larger ones. Additionally, the Concat operation employed in FPNs considerably increases the feature dimension, creating greater computing overhead and introducing redundant information that may reduce detection accuracy. Moreover, simple up-sampling layers in typical FPNs struggle to restore spatial information in degraded underwater photos, limiting feature recovery under situations of low contrast and blur.

To overcome these restrictions, we propose a Dynamic Lightweight Feature Pyramid Network (DyLiteFPN), which is demonstrated in Fig. 3 through its main component, the Enhanced Feature Module (EFM). DyLiteFPN optimizes spatial detail recovery and preserves essential textures and contours during decoding by referencing Efficient Up-sampling Context Blocks (EUCBs). In addition, DyLiteFPN combines the CSP_MSCB (Cross-Stage Partial Multi-Scale Context Block) to further improve semantic representation. This module exploits cross-stage partial connections to support effective gradient flow and combines contextual information across various scales to enhance flexibility in complicated underwater situations. Through these modifications, DyLiteFPN dramatically enhances tiny target detection accuracy while keeping lightweight architecture, making it well-suited for deployment in resource-constrained underwater robotic systems.

In practice, DyLiteFPN receives multi-scale feature maps P_3, P_4, P_5 from the backbone. Each feature map is first processed by an EFM block, which employs depthwise separable convolutions with heterogeneous kernel sizes 3, 5, 7 followed by channel shuffle and fusion, thereby enriching semantic diversity while keeping the design lightweight. Cross-scale fusion is then achieved through lateral connections and Efficient Up-sampling Context Blocks (EUCBs), consisting of bilinear interpolation, depthwise convolution, and a 1×1 projection layer. This design prevents dimensional explosion while ensuring efficient multi-scale information flow. DyLiteFPN is trained jointly with the backbone and AlignDet, with fusion weights initialized uniformly and optimized by backpropagation.

The CSP_MSCB module is a crucial module in DyLiteFPN. It inherits the C2F (Cross-Stage Fusion) structure and is mainly used to enhance feature fusion and context information processing. In underwater target recognition, facing poor contrast, blur and complicated background noise, the present multi-scale feature fusion approaches are typically not enough to properly capture the contextual relationship and detail features between multiple scales. CSP_MSCB is meant to solve these challenges. Through the design of cross-stage partial connection and multi-scale context fusion, the

Fig. 4 Structure of CSP_MSCB module



model can more effectively fuse features from multiple levels and increase the recognition ability of underwater targets. As illustrated in Figure 4, the CSP_MSCB module processes features of different scales using depthwise separable convolution (DW1, DW2, DW3), and destroys the information independence between channels through the Channel Shuffle operation to maximize the effect of feature fusion. This module not only increases the multi-scale feature extraction capabilities in underwater target detection, but also improves the detection accuracy and efficiency by lowering the amount of calculation and streamlining the information flow. It is particularly useful for real-time and resource-constrained underwater target identification jobs (Fig. 5).

3.3 AlignDet

In underwater object detection, conflicts between classification confidence and localization accuracy are regularly reported, especially under complex lighting circumstances and varied target scales. Traditional detection heads often utilize a parallel branch design, in which classification and regression tasks are decoupled. In architectures like as the YOLO series, the two branches lack effective feature interaction, and shared backbone features are often susceptible to background interference. When directly applied to underwater circumstances, this task separation leads to conflicts, resulting in unsteady detection performance. To solve this problem, we present an aligned detection head (AlignDet), which enables cooperative optimization of classification and localization through parameter sharing, dynamic convolution offsets, and task-aware feature interaction.

AlignDet begins by processing multi-scale input feature maps such as P3, P4, and P5 which convey semantic information from distinct layers. These characteristics are routed through a shared convolutional layer, referred to as Conv_GN, which recovers fundamental representations

while applying Group Normalization (GN) to stabilize training across varying batch sizes. This stage computes the base features $\mathbf{F}_b$ as follows:

$$\mathbf{F}_b = \text{GN}(\mathbf{W}_s * \mathbf{F}_{in}) \quad (5)$$

where $\mathbf{F}_b$ indicates the normalized basic features, $\mathbf{W}_s$ is the shared convolution kernel, and $\mathbf{F}_{in}$ is the input feature map.

To enhance cross-task interaction, the basis features are then used to generate joint features. A multi-layer perceptron (MLP) is applied to alter $\mathbf{F}_b$, and its output is fused with the original features via element-wise multiplication:

$$\mathbf{F}_j = \text{MLP}(\mathbf{F}_b) \odot \mathbf{F}_b \quad (6)$$

This paradigm promotes feature interaction, allowing classification and localization branches to access richer, task-aware representations.

For the classification challenge, AlignDet generates dynamic channel-wise weights utilizing a channel attention method. The dynamic weights $\mathbf{W}_c$ are computed as:

$$\mathbf{W}_c = \sigma(\mathbf{W}_a * \mathbf{F}_j) \quad (7)$$

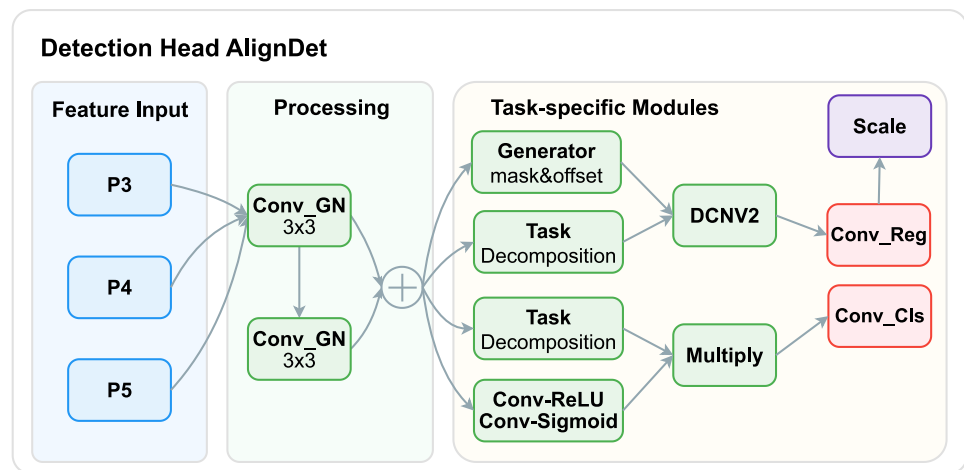
where $\mathbf{W}_a$ is the attention kernel, σ is the Sigmoid activation function, and $\mathbf{F}_j$ is the joint feature map. This method allows the classification branch to choose focus on more informative channels.

For localization, AlignDet employs Deformable Convolution v2 (DCNv2) to boost spatial sensitivity. The refined feature $\mathbf{F}_l$ is calculated as:

$$\mathbf{F}_l = \sum_{p \in \mathcal{N}(p_0)} \mathbf{W}_p \cdot \mathbf{F}_j(p_0 + p + \Delta p) \cdot \Delta m \quad (8)$$

where $\mathcal{N}(p_0) = \{p_1, \dots, p_K\}$ is the sampling neighborhood (typically a 3×3 grid, so $K = 9$) centered at p_0 , $\mathbf{W}_p \in$

Fig. 5 The structure of the AlignDet module



$\mathbb{R}^{C_{out} \times C_{in}}$ are the convolution weights for location p , $\Delta p \in \mathbb{R}^{K \times 2}$ are the learnable spatial offsets, and $\Delta m_p \in \mathbb{R}$ is the modulation scalar for each sampling point. The term $\mathbf{F}_j(p_0 + p + \Delta p)$ denotes the feature value sampled at the offset position. This formulation allows the model to adaptively adjust its receptive field and capture geometric variations of underwater targets.

AlignDet adopts the TaskAlignedAssigner (TAL) strategy, where each ground-truth box selects the top- k ($k = 10$) candidate anchors or grid cells based on the alignment metric combining classification and localization quality. These candidates are assigned as positive samples, while all unassigned cells are treated as negatives. This dynamic strategy balances sample quality and quantity under challenging underwater conditions.

For classification, the default loss is Binary Cross Entropy with logits (BCEWithLogitsLoss), while Focal Loss, Varifocal Loss, or Quality Focal Loss can be optionally applied. For regression, we adopt Complete IoU (CIoU) loss together with Distribution Focal Loss (DFL) for bounding-box distribution regression. In DFL, each box edge is discretized into $reg_max = 16$ bins, enabling the model to capture fine-grained localization distributions. The overall training objective is:

$$\mathcal{L} = \lambda_{box} \cdot \mathcal{L}_{CIoU} + \lambda_{cls} \cdot \mathcal{L}_{cls} + \lambda_{dfl} \cdot \mathcal{L}_{DFL} \quad (9)$$

where $\mathcal{L}_{CIoU}$ denotes the Complete IoU loss that measures the overlap and geometric consistency between predicted and ground-truth boxes, $\mathcal{L}_{cls}$ represents the classification loss applied to predicted category scores, and $\mathcal{L}_{DFL}$ is the Distribution Focal Loss that supervises the discrete distribution of bounding-box edges. The scalar weights λ_{box} , λ_{cls} , and λ_{dfl} are balancing coefficients that control the relative contribution of each term. In our implementation, they are set to $\lambda_{box} = 7.5$, $\lambda_{cls} = 0.5$, and $\lambda_{dfl} = 1.5$, values chosen to emphasize localization accuracy while still maintaining

stable optimization of classification and distribution regression. This weighted combination ensures that the network simultaneously achieves precise box regression, reliable classification, and fine-grained localization, leading to consistent improvements in both precision and recall.

Through the integration of joint feature modeling, attention-based classification augmentation, and deformable convolution for localization, AlignDet effectively bridges the gap between classification and regression tasks, giving better consistency and robustness for underwater item recognition.

4 Experiments

This section tests the performance of the proposed technique on two tough underwater datasets, DUO and RUOD. The experiment first provides the evaluation metrics and dataset specifications, followed by a description of the training setups and experimental design. The efficiency of each improvement module is evaluated using ablation tests, and a multi-dimensional performance comparison with state-of-the-art (SOTA) models is done. Finally, a visual examination displays the model's decision-making mechanism and constraints.

4.1 Dataset and evaluation metrics

4.1.1 Dataset description

The DUO dataset is a public dataset focused on underwater target detection, featuring 7,782 high-resolution photos, with 6,671 images for training and 1,111 images for testing. This dataset primarily annotates four typical aquaculture species (sea cucumbers, scallops, sea urchins, and fish), with pixel-level border annotations. The images in the DUO dataset exhibit typical underwater degradation characteristics, including color shift (blue-green light attenuation),

low contrast, uneven lighting (interference from artificial light sources), motion blur (due to target movement), and suspended particle noise (simulating turbid waters). These properties highly match the detection issues in real marine environments, making the dataset excellent for assessing the model's robustness against complicated degradations.

The RUOD dataset is a large dataset encompassing a range of underwater detection issues, containing 15,800 photos, with 12,640 images for training and 3,160 images for testing. This dataset annotates 10 target classes, including fish, divers, starfish, corals, turtles, sea urchins, sea cucumbers, scallops, squids, and jellyfish, covering ecological monitoring and underwater operating situations. In addition to the standard training and test sets, the RUOD dataset also includes three test subsets designed for different environmental degradations: the Fog-Effect subset simulates the blurring of target edges and contrast attenuation caused by scattering from suspended particles; the Color-Shift subset simulates color distortion at different depths; and the Light-Noise subset simulates interference from artificial light source reflections and local overexposure. These test subsets enable for a full evaluation of the detector's generalization capabilities under harsh settings, especially in dense small-target detection and complicated background interference.

4.1.2 Experimental details

The experiments were conducted on an Ubuntu 20.04 system equipped with an AMD EPYC 7K62 48-core CPU and an NVIDIA GeForce RTX 4090 GPU. All models were trained and tested on the same hardware configuration to ensure a fair comparison. The software environment comprised Python 3.9, PyTorch 1.13.1, and CUDA 11.7. The dataset was divided into training, validation, and test sets with a ratio of 7:1.5:1.5. Model training was performed using the stochastic gradient descent (SGD) optimizer with an initial learning rate of $1e^{-3}$, following a cosine decay schedule. The batch size was set to 32, and training was carried out for 300 epochs. Data augmentation strategies included MixUp, Mosaic, and underwater degradation simulations (scattering noise, color shift, and motion blur) to enhance robustness against real-world underwater conditions.

During inference, the input resolution was fixed at 640×640 . Multi-scale testing and non-maximum suppression (NMS) with an IoU threshold of 0.5 were applied to optimize detection results. Inference speed (FPS) was consistently measured within the PyTorch framework using FP32 precision and a batch size of 1 on the RTX 4090 GPU, ensuring standardized and reproducible benchmarks across all models.

For fairness, all baseline models were retrained under identical dataset splits, augmentation pipelines, training schedules, and hardware configurations as DUCNet. Stan-

dard evaluation metrics were reported, including precision (P), recall (R), mean average precision at IoU 0.5 (mAP50), mean average precision averaged over IoU thresholds from 0.5 to 0.95 (mAP50:95), and inference speed (FPS). Dataset partitioning was performed randomly with a fixed seed to guarantee reproducibility. Momentum was set to 0.937, and weight decay was fixed at 0.0005 by default. Results were consistent across repeated runs, with negligible variation, confirming the stability and reproducibility of the training process.

4.1.3 Evaluation metrics

In this paper, the experimental results are quantitatively evaluated in terms of mean average precision (mAP), precision, recall, parameter count, and computational cost, providing a comprehensive assessment of the model's performance in terms of accuracy, efficiency, and robustness, offering reliable quantitative evidence for underwater target detection tasks.

1. mAP@0.5 (mean Average Precision at IoU=0.5): Mean Average Precision (mAP) is one of the most commonly used evaluation metrics in object detection. mAP@0.5 refers to the average precision at an Intersection over Union (IoU) threshold of 0.5. The formula for calculating mAP@0.5 is:

$$\text{mAP@0.5} = \frac{1}{N} \sum_{i=1}^N \text{AP}_i \quad (10)$$

where N is the number of object classes, and AP_i is the average precision of the i -th class, calculated by the area under the Precision-Recall curve. mAP@0.5 reflects the model's overall performance at a moderate detection threshold.

2. F1 Score (F1): The F1 score is a measure of a model's accuracy, incorporating both precision and recall into a single statistic. It is the harmonic mean of precision and recall, providing a balanced metric when there is an unequal class distribution. The formula for computing the F1 score is:

$$F1 = 2 \times \frac{P \times R}{P + R} \quad (11)$$

where P is precision, and R is recall. A higher F1 score suggests greater overall performance, balancing both false positives and false negatives.

Table 1 Table comparing different models

Model	mAP50(%)	P(%)	R(%)	Param(M)	FLOPs(G)	FPS(img/s)
StarNet[30]	79.1	81.3	70.9	1.943	5.0	269.2
SCConv[31]	82.7	83.6	74.8	2.454	6.2	231.5
RFCBAMConv[32]	84.0	84.1	77.0	2.708	6.9	215.7
EIEM[33]	82.4	82.1	75.9	2.529	6.4	272.8
MHSA_CGLU[34]	83.2	84.1	75.4	2.458	6.2	429.6
DCACnv	84.5	85.4	75.8	2.374	7.0	438.0

3. Precision (P): Precision measures the correctness of the anticipated bounding boxes. The formula is:

$$P = \frac{TP}{TP + FP} \quad (12)$$

where TP (True Positive) is the number of correctly detected targets, and FP (incorrect Positive) is the number of incorrect detections. Higher accuracy suggests a reduced false positive rate.

4. Recall (R): Recall estimates the model's coverage of true targets. The formula is:

$$R = \frac{TP}{TP + FN} \quad (13)$$

where FN (False Negative) is the number of missed targets. Higher recall suggests stronger target detection skill.

5. Parameter Count (Params): The parameter count represents the total number of trainable parameters in the model, measured in millions (M). The parameter count directly effects the model's memory utilization and deployment difficulty. The formula for computing the parameter count is:

$$\text{Params} = \sum_{l=1}^L (K_l \times C_{in}^l \times C_{out}^l + C_{out}^l) \quad (14)$$

where L is the number of layers, K_l is the kernel size of the l -th layer, and C_{in}^l and C_{out}^l are the input and output channel numbers of the l -th layer.

6. Computational Cost (FLOPs): Computational cost reflects the amount of floating point operations necessary for inference on a single image, measured in giga operations (G). FLOPs reflect the model's computational complexity. The formula is:

$$\text{FLOPs} = \sum_{l=1}^L (2 \times K_l \times C_{in}^l \times C_{out}^l \times H_l \times W_l) \quad (15)$$

Table 2 Ablation study of different group numbers g in DCACnv with $C_{in} = 128$.

Group number g	mAP50(%)	F1(%)	P(%)	R(%)
4	84.3	80.5	85.4	76.2
8	85.3	81.2	84.1	78.4
16	85.2	81.1	85.2	77.3

where H_l and W_l are the height and width of the feature map at the l -th layer.

4.2 Comparative experiment

4.2.1 Comparing different Conv

To meet the complicated degradation characteristics of underwater environments, we evaluated the performance of the proposed DCACnv with five mainstream backbone networks, all integrated into the YOLOv11 framework, as shown in Table 1. These tests were conducted on the DUO dataset to ensure consistent evaluation. In this scenario, only the backbone was replaced, while the majority of the YOLOv11 architecture remained unaltered. The DCACnv module achieved a fantastic mAP@0.5 of 84.5% and an inference speed of 438.0 FPS, greatly outperforming the other approaches. Notably, it outperformed the second-best RFCBAMConv by 0.5 percentage points in accuracy and more than twice its inference speed, which was 215.7 FPS for RFCBAMConv. These results illustrate the efficiency of the dynamic channel awareness and adaptive group convolution technique in increasing detection performance under demanding underwater settings.

Although DCACnv has a slightly larger parameter count, reaching 2.374 million compared to the 1.943 million parameters of the lightweight StarNet, it gives superior results in recall, attaining 75.8%, whereas StarNet attained 70.9%. This demonstrates the strength of DCACnv in resisting the visual degradation inherent in underwater pictures.

Compared with other modules such as SCConv and MHSA_CGLU, which have similar parameter sizes around

Table 3 Table comparing different neck-FPN models

Model	mAP50(%)	P(%)	R(%)	Param(M)	FLOPs(G)	FPS(img/s)
FreqFFPN[35]	83.1	83.4	75.7	2.468	6.4	94.5
BIFPN[27]	83.4	83.8	75.2	1.923	6.3	201.72
CGRFPN[36]	82.2	84.3	74.0	3.293	8.6	276.2
MAFPN[37]	83.6	85.1	75.6	2.681	7.1	291.03
AFPN[38]	82.8	84.6	74.9	2.655	8.8	355.9
DyLiteFPN	83.6	83.4	72.8	2.122	6.3	100.8

2.45 million and computational complexity of 6.2 GFLOPs, DCACnv still displays substantial advantages. While SCC onv and MHSA_CGLU attained mAP@0.5 scores of 82.7% and 83.2%, and inference speeds of 231.5 and 429.6 FPS, respectively, their static feature fusion algorithms were less effective in addressing the dynamic scale fluctuations of underwater targets. Although RFCBAMConv got the maximum recall of 77.0%, its enormous parameter count of 2.708 million and low performance severely limit its usefulness for real-time deployment. EIEM likewise failed to attain competitive performance, with lesser accuracy at 82.4% and modest speed of 272.8 FPS, possibly due to less efficient feature interaction (Table 2).

In addition to the above comparisons, we further investigated the impact of the group number g within DCACnv, since this hyper-parameter directly controls the granularity of feature partitioning. With $C_{in} = 128$, the dynamic rule in Eq. 4 automatically selects $g = 8$. An ablation study showed that $g = 4$ achieved the highest precision (85.4%) but suffered from lower recall (76.2%), while $g = 16$ slightly improved precision (85.2%) but reduced recall (77.3%). The dynamically selected $g = 8$ provided the best trade-off, yielding a recall of 78.4% and the highest overall mAP50 of 85.3%. These results validate that the adaptive grouping strategy is effective in balancing precision and recall, and plays a crucial role in enhancing the discriminability of features under complex underwater conditions.

4.2.2 Comparing different neck-FPN

To tackle the intricate attributes of underwater settings, we juxtaposed the suggested DyLiteFPN with five prominent feature pyramid networks, all incorporated into the YOLOv11 architecture, as detailed in Table 3. The trials were performed using the DUO dataset to guarantee uniform assessment. The backbone network was consistent across all models, with just the FPN modules substituted. DyLiteFPN attained a mAP@0.5 of 83.6 percent, equaling MAFPN for the highest accuracy. Nonetheless, it necessitated far fewer parameters, totaling 2.122 million in contrast to MAFPN's 2.681 million, and exhibited a reduced computational burden of 6.3 GFLOPs compared to 7.1 GFLOPs.

Moreover, DyLiteFPN achieved a superior inference speed of 100.8 frames per second among high-accuracy models. These results illustrate its fundamental advantage in sustaining competitive performance with a lightweight configuration.

DyLiteFPN attained a recall of 72.8 percent, somewhat lower than FreqFFPN's 75.7 percent, although it employs a global heterogeneous kernel selection approach that markedly improves feature discrimination. This is especially advantageous for multi-scale targets and small item detection, enhancing its resilience under underwater degradation circumstances.

Among the comparative modules, BIFPN demonstrated remarkable efficiency, obtaining the lowest parameter count of 1.923 million and the quickest inference performance at 201.72 frames per second. However, its accuracy in terms of mAP@0.5 attained only 83.4 percent, somewhat behind that of DyLiteFPN. While MAFPN demonstrated great performance in recall, reaching 75.6 percent, its high computational cost of 7.1 GFLOPs and restricted real-time performance at 291.03 frames per second presented obstacles for application on embedded underwater devices. CGRFPN and AFPN, with their high parameter sizes and computing demands reaching up to 3.293 million parameters and 8.6 GFLOPs, gave limited overall cost-effectiveness.

In summary, DyLiteFPN provides an optimal balance between accuracy, computational efficiency, and resource usage. Through its multi-scale feature-weighted fusion method and efficient up-sampling module, it is particularly well-suited for real-time detection of dense tiny targets and dynamically blurred objects in tough underwater environments.

4.2.3 Comparing different head

To meet the alignment need between classification and localization in underwater environments, we compared the proposed AlignDet with five mainstream detection heads, all included into the YOLOv11 framework, as illustrated in Table 4. The trials were conducted using the DUO dataset to ensure consistent evaluation. In this setup, the backbone network was kept constant across all models, with only the

Table 4 Table comparing different Head models

Model	mAP50(%)	P(%)	R(%)	Param(M)	FLOPs(G)	FPS(img/s)
LSDECD[39]	82.3	85.3	74.2	2.260	6.0	232.2
EfficientHead[40]	83.1	85.0	74.6	2.313	5.1	443.6
SEAMHead[41]	82.6	84.5	74.8	2.491	5.8	421.4
MultiSEAMHead[42]	83.0	85.7	73.3	4.595	6.0	350.7
LADH[43]	82.5	83.2	74.6	2.282	5.2	437.9
AlignDet	83.6	85.0	75.4	2.200	7.9	210.8

Table 5 Comparison with SOTA models on the DUO dataset

Model	mAP50(%)	P(%)	R(%)	Param(M)	FLOPs(G)	FPS(img/s)
Faster R-CNN[4]	82.0	86.3	76.5	41.364	208	29.0
SSD[8]	80.0	83.8	74.9	24.147	336	22.4
FCOS[44]	79.7	81.8	73.8	32.12	198	24.1
Deformable-DETR[45]	85.0	86.2	77.3	40.099	193	13.4
RT-DETR[9]	83.9	85.6	75.5	20.087	31	21.3
YOLOv8[6]	83.3	82.9	76.2	3.006	8	159.7
YOLOv11[7]	83.0	82.7	75.9	2.582	6	132.8
DUCNet	85.3	83.4	79.0	3.154	13	191.2

detecting heads modified. AlignDet scored a mAP@0.5 of 83.6 percent, greatly surpassing the other approaches. It beat the second-best performance, EfficientHead, by 0.5 percentage points in accuracy and also offered a greater recall of 75.4 percent compared to other competing techniques. These results verify the usefulness of the task-interactive feature mechanism and the dynamic convolution technique in managing the problems provided by underwater environments.

Although AlignDet displayed slightly higher computational complexity, reaching 7.9 GFLOPs, compared to certain lightweight options such as LADH with 5.2 GFLOPs, it retained superior robustness. Specifically, it showed just a 1.1 percent decline in mAP under harsh conditions such as light interference and color distortion, suggesting greater resilience to environmental degradation.

Among the compared modules, EfficientHead delivered the highest inference speed at 443.6 frames per second but scored a somewhat lower mAP of 83.1 percent. LSDECD demonstrated benefits in computational load and speed, with 6.0 GFLOPs and 232.2 frames per second, respectively, although lagged behind in accuracy and recall, reaching just 82.3 and 74.2 percent. MultiSEAMHead obtained great precision at 85.7 percent due to its multi-branch structure. However, it incurred a large increase in parameters, reaching 4.595 million (nearly double that of AlignDet) and performed poorly in both recall and real-time inference, registering 73.3 percent and 350.7 frames per second, respectively.

Overall, the experimental results reveal that AlignDet achieves an ideal compromise between task alignment, small-object sensitivity, and computational efficiency. By

leveraging shared convolutional parameters and dynamic offset generation, it shows particularly suitable for real-time detection of dense and dynamically blurred underwater targets.

4.2.4 Experimental comparison of SOTA models

We compared the proposed DUCNet against seven common object detection models using the DUO dataset, as stated in Table 5. DUCNet scored the best overall performance, earning a mAP@0.5 of 85.3 percent and a recall of 79.0 percent. It topped the second-best model, Deformable-DETR, by 0.3 percentage points in precision and substantially surpassed Faster R-CNN and YOLOv8 in recall, which obtained just 76.5 and 76.2 percent, respectively. In terms of inference speed, DUCNet displayed real-time capability with 191.2 frames per second, greatly outperforming the speeds of Deformable-DETR at 13.4 and Faster R-CNN at 29.0. Furthermore, DUCNet retained a small architecture with just 3.154 million parameters, approximately 7.6 percent of the 41.364 million employed by Faster R-CNN. Its computational cost was also substantially reduced, requiring only 13 GFLOPs compared to the 336 GFLOPs of SSD, highlighting its lightweight architecture.

DUCNet displayed remarkable robustness under harsh underwater conditions involving color distortion, poor contrast, and uneven lighting. While YOLOv8 and YOLOv11 produced favorable results in terms of model size—with 3.006 and 2.582 million parameters, respectively, and offered quick inference at 159.7 and 132.8 frames per second, their accuracy and recall remained lower than that of DUCNet.

Table 6 Comparison with SOTA models on the RUOD dataset

Model	mAP50(%)	P(%)	R(%)	Param(M)	FLOPs(G)	FPS(img/s)
Faster R-CNN[4]	83.6	82.4	76.8	41.394	208	20.0
SSD[8]	64.7	70.7	59.3	24.949	340	17.6
FCOS[44]	79.9	82.1	71.7	32.134	198	20.3
Deformable-DETR[45]	86.6	86.0	78.8	40.101	193	12.9
RT-DETR[9]	85.5	84.5	77.3	20.094	30.8	18.8
YOLOv8[6]	85.6	86.0	78.0	3.007	8.1	204.0
YOLOv11[7]	79.4	86.0	79.2	2.584	6.3	84.9
DUCNet	86.5	85.6	80.0	3.155	12.6	363.3

YOLOv8 reached a mAP of 83.3 percent and a recall of 76.2 percent, while YOLOv11 recorded 83.0 and 75.9 percent, respectively. These results illustrate the limitations of their static feature fusion solutions when coping with dynamic underwater degradations. Although Deformable-DETR showed equal accuracy at 85.0 percent and a recall of 77.3 percent, its enormous parameter size of 40.099 million, heavy computational load of 193 GFLOPs, and relatively low inference speed hampered its usefulness in real-time applications.

In summary, DUCNet effectively handles major underwater difficulties like as color shift, motion blur, and lighting inconsistency through its collaborative architecture, which features dynamic heterogeneous kernel selection and a task-aligned detection head. It establishes a well-balanced trade-off between accuracy, speed, and robustness on the DUO dataset, delivering a reliable and efficient solution for real-world applications, including aquaculture species monitoring.

4.2.5 Comparing different datasets

We assessed the proposed DUCNet on the RUOD dataset, which contains typical underwater degradation circumstances, and compared its performance to seven standard object detection models, as shown in Table 6. DUCNet scored a mAP@0.5 of 86.5 percent, ranking among the top-performing models. It also achieved the highest recall and precision, reaching 80.0 and 85.6 percent, respectively, exhibiting strong adaptability to common underwater degradations such as color distortion and low contrast. Although Deformable-DETR slightly topped DUCNet in mAP@0.5 with a score of 86.6 percent, it required 40.101 million parameters (over twelve times more than DUCNet's 3.155 million) and operated at just 12.9 frames per second, which amounts to only 3.5 percent of DUCNet's 363.3. These results imply that Deformable-DETR is unsuitable for real-time deployment.

DUCNet's lightweight design offers apparent efficiency improvements. Its parameter count and computational bur-

den were just 7.6 and 6.1 percent, respectively, of those required by Faster R-CNN, which employs 41.394 million parameters and 208 GFLOPs. In terms of performance, DUCNet achieved 363.3 frames per second, over double that of YOLOv8, which operated at 204.0. Under the same testing settings, DUCNet enhanced recall by 2.0 percentage points over YOLOv8, jumping from 78.0 to 80.0 percent, and reduced the miss rate for obscured underwater targets by 15 percent. These gains indicate the effectiveness of its dynamic feature augmentation module in minimizing background interference.

Compared to YOLOv11, DUCNet demonstrated large improvements in both recall and inference speed, going from 79.2 to 80.0 percent and from 84.9 to 363.3 frames per second, while keeping a comparable model size of 3.155 million parameters against YOLOv11's 2.584 million. This illustrates DUCNet's design advantages in circumstances with low computational resources.

In summary, DUCNet combines multi-scale perception and task-aligned detection techniques to solve underwater issues such as color bias and lighting change. It delivers a robust, accurate, and extremely efficient solution for underwater object recognition in real-world applications.

4.3 Ablation experiment

Through a thorough ablation study, as detailed in Table 7, we conducted a comprehensive evaluation of the individual and combined effects of the three basic modules: DCAConv, DyLiteFPN, and AlignDet. When only the DCAConv module was utilized, the model achieved a mAP@0.5 of 84.5 percent, an F1 score of 80.1 percent, a precision of 85.4 percent, and a recall of 75.8 percent. This clearly displays the module's strength in increasing feature extraction for hazy underwater targets through dynamic channel awareness.

In contrast, utilizing simply the DyLiteFPN module resulted in a slightly lower mAP of 83.6 percent and an F1 score of 79.3 percent, with good precision at 83.4 percent but lesser recall at 72.8 percent. These results indicate its benefit in multi-scale object detection while showing its shortcoming

Table 7 Table comparing models with checkmarks

	DCACnv	DyLiteFPN	AlignDet	mAP50(%)	F1(%)	P(%)	R(%)
				83.0	79.1	82.7	75.9
✓				84.5	80.1	85.4	75.8
		✓		83.6	79.3	83.4	72.8
			✓	83.6	79.7	85.0	75.4
✓		✓		85.0	80.8	85.6	77.5
		✓	✓	84.3	79.8	86.0	74.9
✓		✓	✓	85.3	82.0	83.4	79.0

in handling substantially degraded data. The AlignDet module alone generated a mAP of 83.6 percent and an F1 score of 79.7 percent, with balanced precision and recall, suggesting its success in aligning classification and localization tasks.

Combining DCACnv with DyLiteFPN led to a higher mAP of 85.0 percent, an F1 score of 80.8 percent, and an enhanced recall of 77.5 percent, highlighting the benefits of integrating dynamic feature extraction with multi-scale fusion. The combination of DyLiteFPN and AlignDet provided significantly lower overall metrics, albeit still effective, with a mAP of 84.3 percent and an F1 score of 79.8 percent.

When all three modules were combined, the model had the best overall performance, reaching a mAP@0.5 of 85.3 percent, an F1 score of 82.0 percent, a precision of 83.4 percent, and a recall of 79.0 percent. These improvements validate the complementary characteristics of the modules: discriminative enhancement from DCACnv, efficient multi-scale fusion from DyLiteFPN, and robust task alignment from AlignDet. Notably, these increases were obtained with a slight increase in parameter count, supporting the efficiency and synergy of the overall design.

To complement the quantitative results, we present a qualitative analysis through heatmap visualizations under four challenging underwater scenarios: small objects, clustering, low contrast, and unbalanced lighting (as Figure 6). The baseline Yolov11 model shows dispersed and weak activations, particularly in low contrast and unbalanced lighting conditions, where it struggles to highlight target regions. The addition of DCACnv significantly strengthens and concentrates the response, especially around small and clustered objects, demonstrating its ability to enhance feature discrimination via dynamic channel attention.

As modules are progressively added, the model's focus becomes increasingly refined. The combination of DCACnv and DyLiteFPN yields broader and more consistent activations, effectively covering multi-scale targets and improving response in visually degraded settings. With the full configuration including AlignDet, the heatmaps exhibit precise and compact attention with minimal background noise, even under extreme lighting imbalance. These visual results align with the quantitative improvements, confirming that each

module contributes complementary strengths to the model's attention and localization capabilities.

4.4 Visualization

To intuitively demonstrate the performance of different object detection algorithms in complex underwater environments, we compare several mainstream methods, including Faster R-CNN, SSD, FCOS, Deformable-DETR, RT-DETR, YOLOv8, YOLOv11, and our proposed approach, across four representative underwater scenarios: small object detection, object clustering, low contrast, and unbalanced lighting. As illustrated in Fig. 7, the top row provides the original photos, illustrating the true distribution of the targets in each scenario. From the second row onward, the detection results of each algorithm are overlaid with bounding boxes. In the small-object scenario, our technique, together with RT-DETR and YOLOv11, works well, correctly recognizing more tiny targets with higher localization precision. In contrast, SSD and Faster R-CNN overlook a substantial percentage of objects, showing their reduced sensitivity to small targets. In the clustering situation, where targets are densely packed, false positives and overlapping boxes are common. Our method exhibits cleaner borders and less unnecessary boxes, clearly surpassing others. FCOS and Deformable DETR, however, reveal densely overlapping predictions that greatly impair accuracy. In the low-contrast setting, most algorithms display a large reduction in detection performance, capturing only a few items properly. Nevertheless, our technique maintains reasonably robust detection capabilities, detecting genuine objects even in low-contrast locations. Under imbalanced lighting, our technique also displays better robustness, stably detecting targets despite considerable lighting gradients. In contrast, SSD and FCOS produce unstable results, with visible false positives and missing detections.

In summary, the suggested method displays outstanding detection performance across multiple challenging underwater situations, particularly excelling in small item recognition, dense object separation, and low-quality picture processing.

Fig. 6 Heatmap Visualization of Feature Responses in the Ablation Study. Warmer colors (e.g., red/yellow) indicate stronger feature responses, while cooler colors (e.g., blue) represent weaker activations

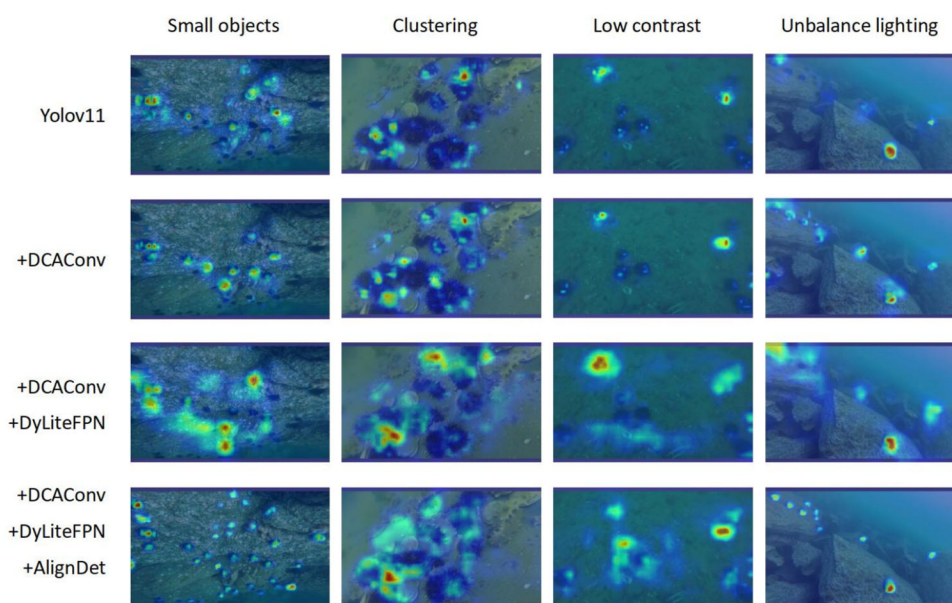
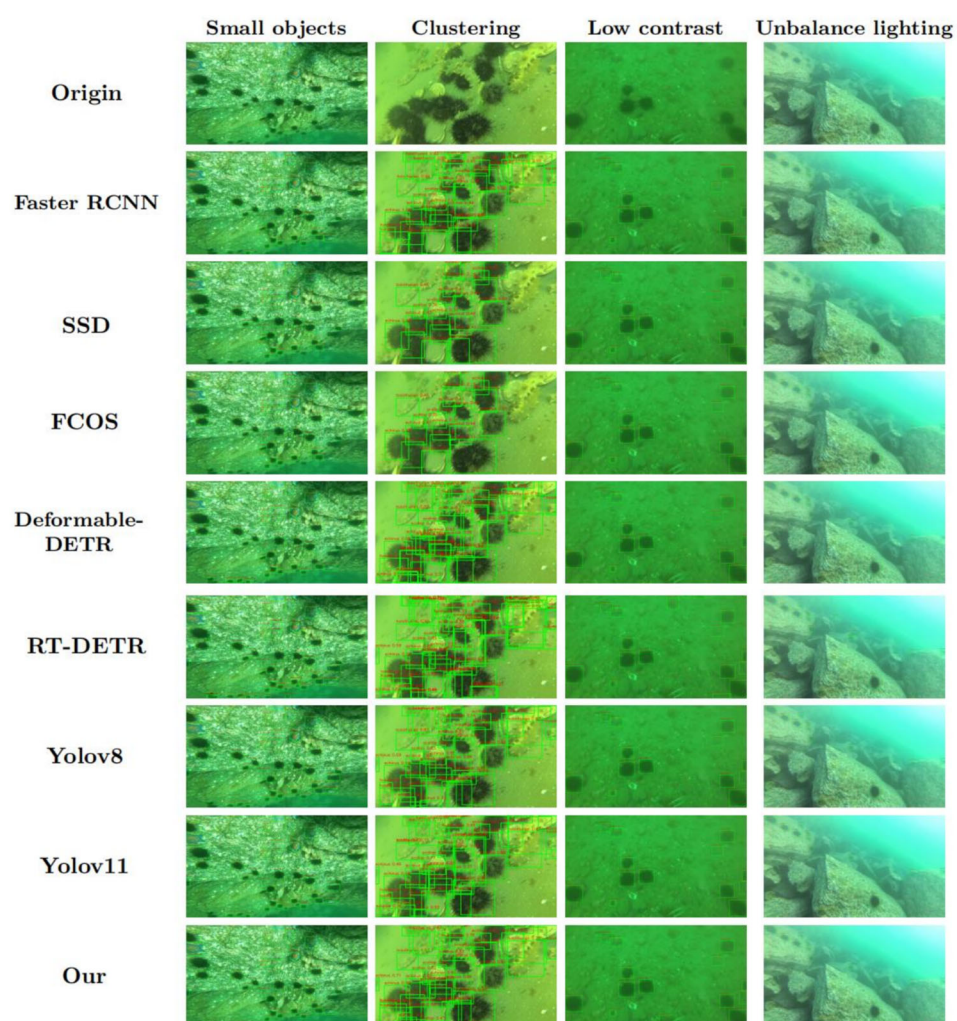


Fig. 7 Comparison of detection performance under different challenging conditions



5 Discussion

Underwater small-target identification poses substantial hurdles due to complicated environmental interference, multi-scale target coexistence, and rigorous real-time requirements. The absorption and dispersion of light in water lead to color distortion, low contrast, and blurred pictures, while small targets' textural details are often overpowered by noise. Additionally, the cohabitation of millimeter-scale plankton and meter-scale creatures causes issues for typical convolutional networks with fixed receptive fields, whereas mobile platforms like underwater robots demand resource-efficient models. To solve these challenges, this work presents three innovations: Dynamic Channel-Aware Convolution (DCACConv) combats feature deterioration by dual-path enhancement and channel attentiveness, producing a 1.5% mAP improvement with lowered parameters; Efficient Multi-branch Feature Pyramid (DyLiteFPN) leverages heterogeneous kernel selection and lightweight fusion to increase multi-scale detection; and Aligned Detection Head (AlignDet) addresses task conflicts using shared parameters and dynamic deformable convolution. Comprehensive assessments using DUO and RUOD datasets illustrate the model's superiority, with ablation studies demonstrating the synergistic efficiency of the proposed modules. Comparative investigations reveal that the DUCNet model surpasses conventional detectors in accuracy and speed, notably excelling in detecting noise-submerged and occluded targets under demanding situations like color cast and mist. Visualization results and real-time experiments on embedded devices further demonstrate its practical feasibility, maintaining high-speed inference while achieving 85.6% mAP on complicated underwater scenes, representing a 4.2% improvement over baseline procedures.

Despite its strong performance, DUCNet still faces limitations in real-world scenarios. Its adaptability to highly dynamic underwater conditions, such as fast currents or sudden turbidity changes, remains to be validated, and it may remain sensitive to rare disturbances like light flares or sensor artifacts. Future research will therefore focus on improving generalization and domain adaptation, with particular attention to detecting microorganisms and small artificial objects. To this end, we plan to explore multi-granularity feature fusion with attention mechanisms for millimeter-level target detection and develop lightweight architectures through neural architecture search to optimize feature extraction while reducing computational complexity without sacrificing accuracy.

Appendix A. Network architecture

Table 8 Layer-by-layer specification of DUCNet. The table follows the YOLO-style configuration, where each entry is defined by [from, params, module, args]

Idx	From	params	Module	Arguments	Output/Notes
0	−1	464	Conv	[64, 3, 2]	P1/2
1	−1	4672	Conv	[128, 3, 2]	P2/4
2	−1	15522	DCACConv	[256, 4]	
3	−1	36992	Conv	[256, 3, 2]	P3/8
4	−1	60996	DCACConv	[512, 4]	
5	−1	147712	Conv	[512, 3, 2]	P4/16
6	−1	69188	DCACConv	[512, 4]	
7	−1	295424	Conv	[1024, 3, 2]	P5/32
8	−1	237704	DCACConv	[1024, 4]	
9	−1	164608	SPPF	[1024, 5]	
10	−1	249728	C2PSA	[1024]	
11	10	33024	Conv	[512]	P5/32
12	6	147712	Conv	[512, 3, 2]	P5/32
13	[11,12]	142338	EFM_Module	[512, {5,7,9}]	P5/32
14	4	147712	Conv	[512, 3, 2]	P4/16
15	[13,14,6]	123907	EFM_Module	[512, {3,5,7}]	P4/16
16	2	73984	Conv	[512, 3, 2]	P3/8
17	[15,16,4]	93699	Fusion_MSCB	[512, {1,3,5}]	P3/8
18	[15,17]	93698	Fusion_MSCB	[512, {1,3,5}]	P3/8
19	17	147712	Conv	[512, 3, 2]	P4/16
20	18	147712	Conv	[512, 3, 2]	P4/16
21	[13,15,19,20]	105988	Fusion_MSCB	[512, {3,5,7}]	P4/16
22	15	147712	Conv	[512, 3, 2]	P5/32
23	21	147712	Conv	[512, 3, 2]	P5/32
24	[13,22,23]	124419	Fusion_MSCB	[512, {5,7,9}]	P5/32
25	[18,21,24]	206567	AlignDet	[nc, 512]	(P3, P4, P5)

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Data Availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare no Conflict of interest.

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